



Evaluating Ongoing Decision Support System:

A Case Study

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ABSTRACT

The aim of this paper is to put into practice a framework we proposed in an earlier work for evaluating on-going decision support systems. The framework serves as a base for a method to compute evaluation criteria scores using an AHP model. The paper also illustrates the automation of this method into a tool to enable GASCO company to choose between two already implemented DSSs in order to reduce IT operating cost. The evaluation covers three axes: Process, outcome and decision maker metrics. The results highlight the importance of using users' learning as a measure to determine DSS success.

KEYWORDS

Decision support systems, software evaluation, user learning

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1 Introduction

Decision support systems (DSS) represent and process knowledge in ways allowing decision making to be more productive and to improve decision quality and outcome [1]. They provide information gathered internally and from external sources along with recommendations and guidance to the decision makers.

With constant emerging organizational and business needs, Investing in Decision support systems comes with high costs and risks. As improved decision performance outcomes are a result of appropriate combinations of individual, task and technology factors [2]. A comprehensive understanding of effective decision support requires consideration of technology design, as well as the fit between the individual and the task. Many research works address the question of DSS evaluation, but few are specific to the post implementation phase while still providing a wholistic approach to the question of evaluation [3]. In this sense, a global evaluation of ongoing DSS must consider decision value metrics (covering process and outcome) and decision maker related metrics (based on the effects of DSS on users).

In an earlier work we proposed a framework [4] by incorporating learning as an ongoing DSS measure in the Rao and Rhee framework [5]. In the following study, we use this framework to build an AHP (Analytical hierarchy process) based method in order to identify, allocate weights and give scores to a set of evaluation criteria for ongoing DSSs.

As an attempt to adapt and automate this method for GASCO, an oil and gas company, the study also presents the EvaDSys tool that was developed to support a group of evaluators with a decision regarding two ongoing decision support systems namely GASFIN and PLAFIN DSSs. This decision will help the company making an investment choice between GASFIN and PLAFIN already implemented, but one only had to be kept for the sake of budget optimization. The evaluation committee was involved in crafting the measure selection process and in weighting the criteria. The evaluators gave scores for some of the criteria while others were calculated automatically and made available for EvaDSys for computing.

The remainder of the paper is organized as follows: In the second section we discuss the proposed evaluation framework, and how it can be applied. Section 3 presents the method and the evaluation hierarchy. Section 4 will be dedicated to the evaluation of on-going DSSs in GASCO and will describe the adapted hierarchy and the built tool and its design features and application, followed by a discussion and a conclusion.

2 A Framework for On-going DSS Evaluation

In order to determine ongoing DSSs' success, the evaluation process should consider their contributions to improving the process of and the outcome from the decision making, along with their impact on the decision maker. In the light of the proposed model by Forgionne [6], validated through several experiments (Healthcare, small business...) and findings of the research model, it has been demonstrated that for any successful global multi-criteria evaluation of ongoing DSS, we should involve two important components: decision value and decision maker [7]. The decision value is issued from the combination of both the decision outcome and the decision-making process efficiency [8]. The decision maker component has been relying mostly on satisfaction metrics and was subject to higher subjectivity and difficult to quantify or compute.

We proposed an extended evaluation framework [7] that incorporates learning metric to the decision maker component in order to overcome the subjectivity issue, and hence produce four measures of evaluation: Quality, efficiency, satisfaction and learning.

The resulting framework is illustrated in Figure1. Decisional guidance (criteria) is used to evaluate decision outcome on quality, decision-making process on efficiency and decision maker experience on satisfaction and learning.

Decisional guidance can provide the steps of the evaluation and the criteria through the implementation of the typology discussed earlier: Target representing system capabilities, Directivity and recommendations provided to the use, the Mode and user involvement in the guidance, decision process duration, interface formats, Audience adaptation...

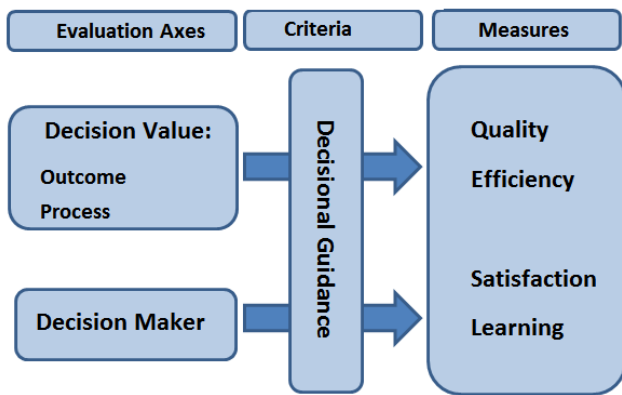


Figure 1: The proposed framework

This framework can be widely used over various types of ongoing DSSs for a range of combinations of domain and technology. The evaluation is based on four measures: decision quality, efficiency of decision-making process, decision-maker's satisfaction and user's learning. This approach helps the interested parties involved in the

evaluation process to draw a big picture associated with the evaluation of their own domain-technology-specific DSS. Moreover, it can create the link between design requirements and evaluation process and then be used to assess the outcomes of DSS development and usability. Multiple-criteria DSS evaluations can be useful in conducting a reconciliation and in providing a consistent basis for future studies on DSS effectiveness.

3 An Evaluation Method for On-going DSS

The method we propose is based on AHP (Analytical Hierarchy process).

3.1 Analytical Hierarchy Process

AHP [9][10][11] is a multi-criteria model that allows a comparison of alternatives by structuring criteria into a hierarchy. It is an approach that derives ratio scales from both discrete and continuous paired comparisons that are used to rank the alternatives. Alternatives are compared each of the criterion defined by the user/evaluator, the AHP takes in charge synthesizing the results following the hierarchy to produce a score for each alternative.

The process of alternatives' evaluation follows several steps; first, the evaluator has to define the importance level of all the criteria, after that, the alternatives are scored against each criterion. After computing all scores, the process will be able to determine the best alternative, given the weights and scores by criterion/alternative. AHP has been widely used in supporting decision making and evaluating intelligent systems [12].

3.2 Criteria Definition

The evaluation hierarchy uses the main axis of the framework presented above and defines eligible measures discussed in our literature review [5][6], to build a four-level hierarchy model composed of criteria and sub-criteria (Figure2).

The two main evaluation axes are: Decision value and decision maker and shall be expressed using criteria extracted from decision guidance elements to measure quality and efficiency of both the process and outcome and quantify satisfaction and learning improvement of decision makers.

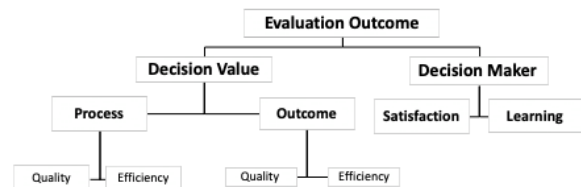


Figure 2: The evaluation hierarchy

DSS are used to improve the decision process and outcome, this can be expressed as improved profit, reduced cost or improved

information accuracy. Decisional systems improved to enable faster decision making, supporting fast changing data/environments, allowing collaboration and providing reliable information, all while keeping users satisfied and allowing them to be better decision makers. The criteria hierarchy must take into consideration all of the above. Decision value is a combination of process efficiency and outcome. Process efficiency can be evaluated on two major axes [ref]: Analysis tools used to process data and responsible for computing results, and design features illustrated by user interface elements, ease of use and usability rates.

In another hand outcome focuses on the benefits gained by the organization from using the system, and are illustrated by improved process (such as improved decisions processing time, and decisions accuracy), the gain in term of budget and return on investment, and finally Organization overall performance in term of resources and budget utilization specific to the context of DSS utilization.

On the basis of a literature review, we presented in an earlier work [3][7] a collection of DSS evaluation criteria proposed and validated by several studies. We then express quality and efficiency in quantifiable criteria needed to measure DSS success.

For the decision maker part of the hierarchy, satisfaction has been highly cited as an evaluation criterion, mostly measured via questionnaires addressing psycho-cognitive effects of DSS on users such as regret avoidance, persuasiveness... Learning is the measure we proposed earlier [4] and is based on usage data and user's behavioral aspects captured during the interaction with the system. The hierarchy branches under process and outcome nodes are presented in Table 1.

Table 1: DSS evaluation criteria definition

Decision Value metrics				
Process		Outcome		
Analysis Tools	Design Features	Improved process	Organization performance	Budget optimization
Data Processing Knowledge Processing Task Complexity Architectural capability Evolving Data quality Adequate processing Interoperability Flexibility Stability Sustainability Value of Processed information Process time	Information presentation Usability Interestingness Service-task User experience Low Cognitive load User participation	User performance Regret avoidance Decision time Accuracy Reduced complexity Validity Constructiveness Context awareness Decision time	Adequate responses Scalability Organization performance Social influence Utilization Resource optimization	Increased growth Return on investment Invested budget vs Gained budget

The resulted hierarchy can be adapted to any organizational context or domain-specific systems by allowing the evaluators to select/remove criteria that do not serve their evaluation process.

4 A case study: Evaluating On-going DSSs in GASCO

4.1 Organizational Context: GASCO

The organization where the case study has been conducted is an Oil and Gas company. GASCO processes gas from onshore and offshore sources and produces a range of products including methane, ethane, propane, butane, paraffinic naphtha, condensates and Sulphur. Every day GASCO delivers over 10,500 tons of ethane, in addition to 4,300 MMSCF of sales gas, 217,000 barrels of condensate and 33,900 tons of propane, butane and naphtha. These products fuel UAE' industries, supporting the manufacture of aluminum, cement, steel, fertilizer, chemical feedstock, water and electricity.

The extraction plants are located in western region of Abu Dhabi and Ruwais and are connected to as long as 3000 KMs of pipeline network aiming to distribute gas and other products worldwide.

GASCO works closely with other ADNOC Group companies, to supply feed gas, to process the condensate, to manufacture plastics using ethane and to export the products internationally. The IT department of GASCO manages all the organization's needs in terms of technological support. The inventory counts a considerable amount of well-known software and hardware products. These tools assist the company in all the stages of industrial production and internal management. Decision support systems are used across departments and are assisting with financial/budgeting, human resources and projects management decisions.

In recent years, many budget' optimization campaigns had been launched in response to the financial instability led by the continuous changes in oil prices. In this perspective, the organization board had decided to lower the budget allocations for all departments including the IT. One way to limit the costs for IT management was to reduce the number of licenses and dispose of redundant applications.

It is in this perspective that an objective evaluation of existing decision support systems became a necessity.

4.2 The Evaluation Hierarchy

A focus group of evaluators representing the different evaluation constituencies (users, development team, purchasing...) was gathered to decide on the evaluation hierarchy. The discussion was conducted to retain the evaluation metrics from the ones presented in Table I according to the needs expressed by the organization members. The resulting hierarchy is illustrated in the Figure.3.

4.3 The EvadSys: A System for On-going DSS Evaluation

The weights used for weighting the hierarch criterion are specific to the evaluator, since different users may provide different values based of their environment understanding and perception of quality [12]. The automation of the method will provide the possibility to gather inputs from multiple users and consolidate them for computing the scores. EvaDsys system was developed to automate

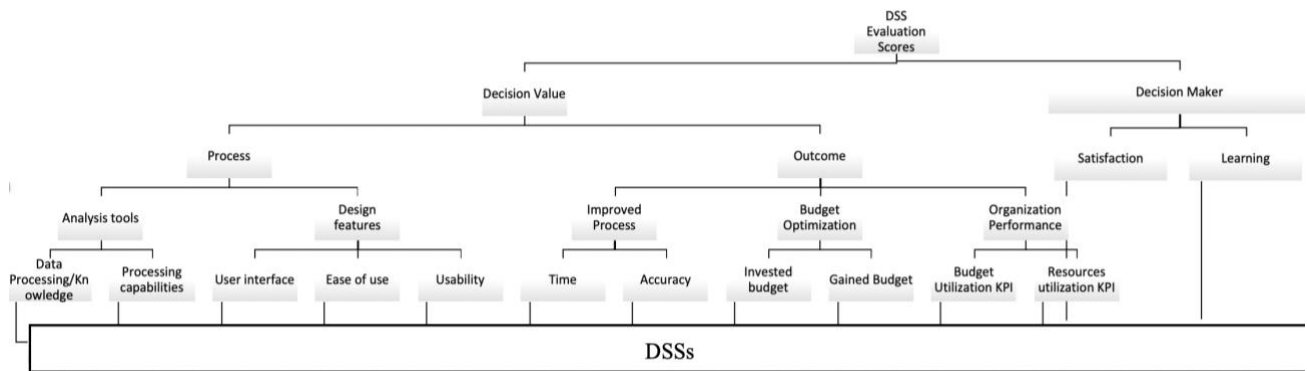


Figure 3: The evaluation hierarchy for GASCO

the method described earlier in order to enable decision makers in GASCO to evaluate ongoing DSS. To keep the alignment with the user requirements and to simplify the calculations, it has been convened that the system will prompt users to enter criteria weights in the form of sliders instead of using the AHP based on user judgments.

The system should ensure the normalization of the weights at each level (The sum of weights at every level should be equal to 1) and should construct the final weights by averaging the collected values from all users.

The rest of the functional and technical requirements are summarized in the following points:

DESIGN FEATURES

- Ease of use
- Provide interfaces for criteria definition, weights allocation and scoring
- Possibility to create/edit hierarchies
- Provide appropriate tools for weights and scoring other than numerical (Sliders)
- Possibility to use the same hierarchy to evaluate different DSSs.
- Possibility to compare scores across users and evaluated systems

TECHNICAL FEATURES

- Compute scores based on defined hierarchy
- Normalize weights in all levels
- Consume data from user inputs and extra sources
- Data historization

DATA SOURCES

- Learning scores collected from DSSs usage data
- Collect KPIs values from different systems' databases
- EvadSys's own database for user defined scores

USER PROFILES

- Admin role: create hierarchy, define data sources, launch calculations plus the normal user profile responsibilities.
- Regular User role: Define weights, define scores, consult results via dashboards.

4.3.1 Criteria and Weight Definition. In the Admin panel the system allows the user to create criteria at any branch of the hierarchy. After that, the users can access the system and define the weight (Figure. 4). The user chooses the parent leaf of the hierarchy (from a drop-down list) to which the criterion will be attributed. (Figure. 5).

For the weights, in order to avoid the numerical scales, we provided a bi-polar slider. The user slides the arrow from irrelevant to high importance. (Figure. 6)

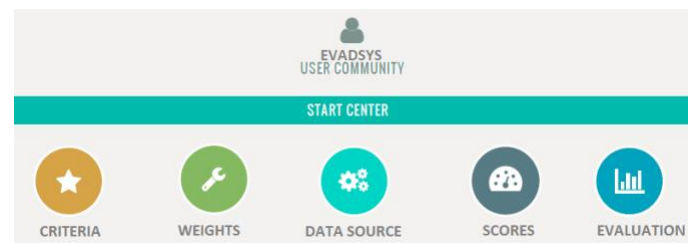


Figure 4. EvadSys Home screen



Figure 5. Criteria definition in EvadSys

For the scores, the evaluation process uses two different sources for the scores:

Figure 6. Weight allocation in EvadSys

Database

The database contains the outcome of several processes running to capture users' learning and satisfaction, and the values of other KPIs serving as criteria to the evaluation. These KPIs represent pertinent organization and process performance metrics and include relevant personal efficiency and productivity results, along with organization's achievements in terms of target reaching, resource utilization and budget optimization (Table 2).

Learning is calculated based on a set of internal processes capturing user's interaction with the system represented by memory, processing speed, awareness and other cognitive capabilities. This process has been detailed in one of our previously published work [4]. Satisfaction is a result of periodic surveys that the DSS suggests to users upon completing their process tasks.

The rest of the database-sourced measures are KPIs calculated automatically by the system pulled from corporate financial dashboard and usage data.

Evaluators Input

The system also uses a screen for users to insert scores of some evaluation criteria. These measures are subjective and represent the evaluators' opinions and how they perceive the system. They cover the user-related components, like design features, ease of use, accuracy of decisions and they belong to the lowest level of the hierarchy (Figure 7).

Table 2: Database Sourced Scores

Criteria	Source	Details
Learning	Computed learning scores	From the learning process
Satisfaction	Users' Satisfaction survey	From the satisfaction process
Invested Budget	DSS Database	From IT purchasing database
Gained Budget	DSS Database	Ratio of budget gained after using the DSS
Budget utilization KPI	DSS Database	Budget conformity planned against forecast
Resources utilization KPI	DSS Database	Time spent on the task by users

Fig. 7. User defined scores interface

4.4 Evaluation of GASFIN and PLAFIN

In the following, we use EvadSys, particularly, to evaluate and compare two DSSs owned by GASCO.

4.3.2 PLAFIN and GASFIN DSSs: PLAFIN and GASFIN are two decision support systems implemented to manage the financial decisions. They are used by different divisions and users and help achieving the same objectives; that is assisting users with budget allocations, future investments and consolidating financial data for higher management approvals and follow ups.

GASFIN: Is a DSS based on Hyperion and BPM application. Data interchanged by the two systems are then stored in GASFIN database to build custom BI reports and provide data insights. It presents executive summaries, runs comparison scenarios and provides predictive models and suggestive answers. The system also provides external educative links, usage history and tutorials. GASFIN is used by Operations division to run budget allocations and consolidate inputs from different departments to the BPM layer.

PLAFIN: Is a DSS based directly on the EBS ERP and K2 Platform, equipped with predictive analytics and heuristic models to determine investment scenarios and support expenditure patterns for on-going and future projects. It is used by projects division to optimize budget allocating reduce the risks of overspending. It has been equipped with different decision aids also such as elaborated dashboards, history and information materials.

4.3.3 Computing Scores: The evaluation of both PLAFIN AND GASFIN systems aims to measure their success, and their impact on the organization and decision makers. Without neglecting the opinion of each of the system's users, we used EvaDSys to make it possible for the evaluators to choose between the two systems. The approach used allows the evaluator, to build a hierarchy, and to allocate weights and compute scores for each criterion, in the perspective of a holistic and global evaluation. It produces not only a single measure for the whole evaluation process, but it permits the scores' calculation for the criteria, and to compare it with the results of the other system, following the steps described previously.

The values are exhibited in table 3.

Table 3: Weights and Scores computed by EvadSys

Level		Criteria	Weights	Scores
I	1	Decision Value	0.55	0.5982 / 0.4018
	2	Decision	0.45	0.5780/ 0.4220
	1.1	Process	0.65	0.5 / 0.4234

II	1.2	Outcome	0.35	0.7805 /0.3616
	2.1	Satisfaction	0.38	0.4300/ 0.4651
	2.2	Learning	0.62	0.6527/ 0.4170
III	1.1.1	Analysis Tools	0.57	0.6572/0.5921
	1.1.2	Design	0.43	0.2916/0.1997
	1.2.1	Improved	0.22	0.3162/ 0.2118
	1.2.2	Budget	0.47	0.4561/0.4822
	1.2.3	Organization	0.31	0.4965/0.2109
IV	1.1.1.1	Data Processing/	0.68	0.7123/0.647
	1.1.1.2	Processing	0.22	0.7856/0.6915
	1.1.2.1	User interface	0.18	0.2269/ 0.2
	1.1.2.2	Ease of use	0.7	0.2325/0.1081
	1.1.2.3	Usability	0.12	0.7334/0.7335
	1.2.1.1	Time	0.58	0.45/0.3
	1.2.1.2	Accuracy	0.42	0.1314/0.09
	1.2.2.1	Invested	0.34	0.61/0.58
	1.2.2.2	Gained budget	0.66	0.3768/0.43
	1.2.3.1	Budget	0.5	0.72/0.12
	1.2.3.2	Resources	0.5	0.273/0.3018

On the basis of the weights established above, the final scores for the two DSSs were calculated: PLAFIN (0.589) and GASFIN (0.4108).

4.5 Main Results Analysis

The results show that the PLAFIN DSS is scoring better than GASFIN in most areas of evaluation. It has got a higher overall score and maintained a distinct positive performance in most areas of evaluation over its rival.

While we remark that in terms of decision maker point of view, the difference is prominent due to the introduction of learning as a metric. Limiting the evaluation to user satisfaction only would have produced a small difference in score, making it a very tied evaluation. The Learning scores drawn from usage data independently from user inputs have produced an objective and reliable metric to determine the nature of usage relationship between the decision makers and the DSS.

The learning metric has given a solid base for decision-maker oriented scores and provided conclusive values of the utility of the systems and elevated the difference between scores to 0.156 in favor of PLAFIN.

5 Conclusion

The goal of this paper was to present our method of ongoing DSS evaluation based on a framework that we proposed in an earlier research work. The method consists of the definition of an AHP hierarchy, with weighted criteria referenced in the literature. The

method was validated by the mean of a case study in an oil and gas company called GASCO.

For the sake of evaluating two DSSs owned and operated by GASCO, we developed EvaDsys system to help the company chose the system worth keeping as a strategic investment. The results of the evaluation of both PLAFIN and GASFIN DSSs has highlighted the importance of using captured learning scores as an indicator of decision maker interaction with DSS rather than limiting this decision maker evaluation perspective to user satisfaction.

REFERENCES

- [1] Holsapple, C. W. "Decisions and Knowledge", 2008. In: Handbook on Decision Support Systems 1. International Handbooks Information System. Springer, Berlin, Heidelberg, 2008.
- [2] Parkes, A. "Designing effective decision support using decisional guidance", 2010 PACIS, 91.
- [3] Boukhayma K., Elmanouar, A., 'Evaluating decision support systems: a literature review', 2015. In Proceeding of ISDA, Marrakech 2015. Paper 88.
- [4] Boukhayma K., Elmanouar, A., "Evaluating Decision Support Systems' Effect on User Learning: An Exploratory Study", 2018. In: Rocha Á., Adeli H., Reis L., Costanzo S. (eds) Trends and Advances in Information Systems and Technologies. WorldCIST'18 2018. Advances in Intelligent Systems and Computing, vol 746. Springer, Cham.
- [5] Rhee C., Rao H. R., Evaluation of Decision Support Systems. In: Handbook on Decision Support Systems 2. 2008. International Handbooks Information System. Springer, Berlin, Heidelberg.
- [6] Forgionne, G. "Decision-Making Support Systems Effectiveness: The Process to Outcome Link". 2000. Information Knowledge-Systems Management 2.
- [7] Boukhayma, K., Elmanouar, A. "Towards a global DSS evaluation method using decisional guidance". Journal of Scientific & Engineering Research Volume 8, Issue 8, 2017
- [8] Forgionne, G., Mora M., Gupta J., Gelman O. "Decision-Making Support Systems", 2005. Encyclopedia of Information Science and Technology, Hershey, PA, Idea Group.
- [9] Parikh M., Fazlollahi B., Verma S. "The effectiveness of decisional guidance: an empirical evaluation", 2015. Decision Sciences 32 (2).
- [10] Rai, A., Lang, S. S., Welker, R. B., (2002). Assessing the validity of IS success models: An empirical test and theoretical analysis. information systems research, 13 (1).
- [11] Reason, J., 2008. The Human Contribution: Unsafe Acts, Accidents and Heroic Recoveries. Surrey, England, Ashgate.
- [12] Henry, W. W., Potts, c., Jeremy, C., Wyatt Douglas, G., Altman, 2001. Challenges in Evaluating Complex Decision Support Systems: Lessons from Design-a-Trial. In Proceedings of 8th Conference on Artificial Intelligence in Medicine in Europe, 2001.
- [13] Moreau, É. M. F., 2016. The impact of intelligent decision support systems on intellectual task success: An empirical investigation. Decision Support Systems (42:2) (2016).
- [14] Chen, Z., 1992. User responsibility and exception handling in decision support systems. Decision Support Systems. 8(6), 537–5